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1-1 PRIVATE TUTORS

ALGEBRA

# Linear Inequalities

Name _____ Date _____ No calculator Version 1

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**KEY WORDS** inequality *(whārite-kore)* · at least · at most · number line *(rārangi tau)*

## A Reading and writing inequalities ●○○

**Remember:** $<$ less than, $>$ greater than, $\leq$ less than or equal to, $\geq$ greater than or equal to.

**1.** Write each sentence as an inequality.

**a)** $x$ is greater than 5 _____

**b)** $x$ is at most 7 _____

**c)** $x$ is at least −2 _____

**d)** $x$ is below 10 _____

**e)** $x$ is no more than 4 _____

**f)** $x$ is more than −3 _____

**2.** Circle every number that makes the inequality true.

**a)** $x > 3$: $1$, $3$, $3.5$, $8$

**b)** $x \leq -2$: $-5$, $-2$, $0$, $2$

**c)** $x \geq 0$: $-1$, $0$, $0.1$, $6$

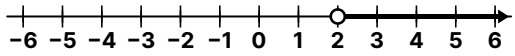
**d)** $x < -4$: $-7$, $-4.5$, $-4$, $4$

## B Number lines ●○○

**Remember:** An open circle means the number is not included. A filled circle means it is included.

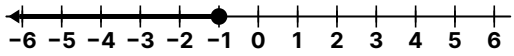
**3.** Write the inequality shown on each number line.

**a)**



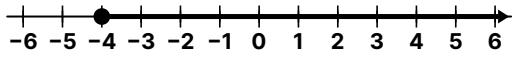

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**b)**



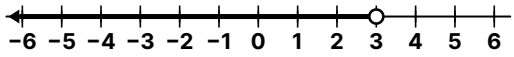

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**c)**




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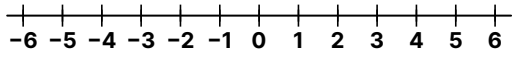
**d)**



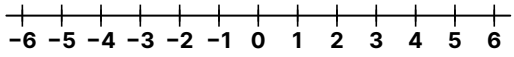

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**4.** Show each inequality on its number line.

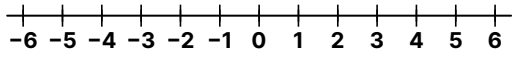
**a)** $x \geq 1$



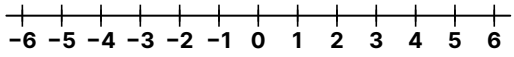

**b)** $x < -2$




**c)** $x > -5$




**d)** $x \leq 4$






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## C Solving inequalities ●●○

**Remember:** Solve an inequality the same way as an equation. Keep the inequality sign in every line.

**5.** Solve.

**a)** $x + 3 > 7$ **b)** $2x \leq 10$ **c)** $x - 6 < -8$

**d)** $3x - 1 \geq 11$ **e)** $5x + 2 > 17$ **f)** $4x - 7 \leq 9$

**g)** $\frac{x}{2} < 4$ **h)** $2x + 9 \geq 3$ **i)** $\frac{x}{3} + 1 > 4$

**j)** $2(x + 3) > 10$ **k)** $3(x - 2) \leq 9$ **l)** $5(2x + 1) < 15$

## D Dividing by a negative ●●●

**Watch out:** When you multiply or divide both sides by a negative number, turn the inequality sign around.

**6.** Solve.

**a)** $-2x > 6$ **b)** $-x \leq -4$ **c)** $-3x + 1 < 10$

**d)** $5 - x \leq 2$ **e)** $10 - 2x > 4$ **f)** $-4x - 3 \geq 5$

**g)** $7 - 3x < 13$ **h)** $-5x + 2 \leq -8$ **i)** $1 - 4x \geq 13$

## E Both sides and fractions ●●●

**7.** Solve.

**a)** $5x - 3 \geq 2x + 9$ **b)** $7x + 2 < 3x - 6$ **c)** $4x + 5 > x + 14$

**d)** $2x + 9 \leq 5x + 3$ **e)** $\frac{x + 1}{2} < 4$ **f)** $\frac{2x - 1}{3} \geq 3$

**g)** $\frac{3x + 4}{2} > 5$ **h)** $3x - 8 > 6x + 4$ **i)** $\frac{5x - 2}{4} \leq 7$

## ★ Extension

**8.** Solve $-3 < 2x + 1 \leq 9$. Then list the whole numbers that satisfy it.

**9.** Movie tickets cost \$14 each and a large popcorn costs \$8. Mere has \$50 and buys one popcorn. Write an inequality for the number of tickets $x$ she can buy, and solve it. What is the largest number of tickets?


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# Linear Inequalities

Name _____ Date _____ No calculator Version 2

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**KEY WORDS** inequality *(whārite-kore)* · at least · at most · number line *(rārangi tau)*

## A Reading and writing inequalities ●○○

**Remember:** $<$ less than, $>$ greater than, $\leq$ less than or equal to, $\geq$ greater than or equal to.

**1.** Write each sentence as an inequality.

**a)** $x$ is greater than 8 _____

**b)** $x$ is at most 3 _____

**c)** $x$ is at least $-4$ _____

**d)** $x$ is below 6 _____

**e)** $x$ is no more than 9 _____

**f)** $x$ is more than $-1$ _____

**2.** Circle every number that makes the inequality true.

**a)** $x > 2$: $0$, $2$, $2.5$, $9$

**b)** $x \leq -3$: $-6$, $-3$, $0$, $3$

**c)** $x \geq 1$: $-1$, $0.9$, $1$, $4$

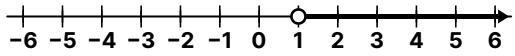
**d)** $x < -1$: $-3$, $-1.5$, $-1$, $1$

## B Number lines ●○○

**Remember:** An open circle means the number is not included. A filled circle means it is included.

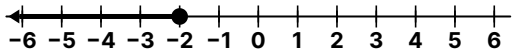
**3.** Write the inequality shown on each number line.

**a)**



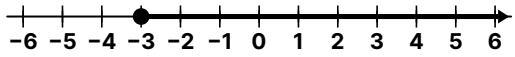

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**b)**



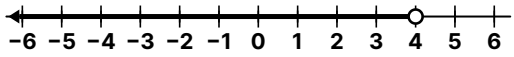

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**c)**




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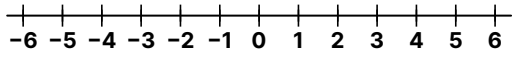
**d)**



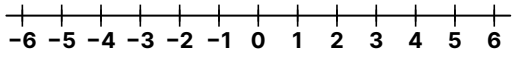

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**4.** Show each inequality on its number line.

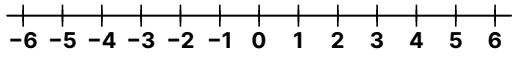
**a)** $x \geq 2$



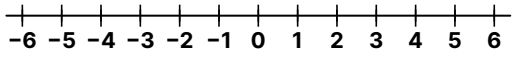

**b)** $x < -1$




**c)** $x > -4$




**d)** $x \leq 5$






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## C Solving inequalities ●●○

**Remember:** Solve an inequality the same way as an equation. Keep the inequality sign in every line.

**5.** Solve.

**a)** $x + 5 > 8$ **b)** $3x \leq 12$ **c)** $x - 4 < -7$

**d)** $2x - 3 \geq 7$ **e)** $4x + 3 > 11$ **f)** $5x - 6 \leq 9$

**g)** $\frac{x}{3} < 4$ **h)** $2x + 11 \geq 3$ **i)** $\frac{x}{4} + 2 > 4$

**j)** $3(x + 2) > 12$ **k)** $2(x - 5) \leq 6$ **l)** $4(2x + 1) < 12$

## D Dividing by a negative ●●●

**Watch out:** When you multiply or divide both sides by a negative number, turn the inequality sign around.

**6.** Solve.

**a)** $-3x > 6$ **b)** $-x \leq -6$ **c)** $-2x + 3 < 11$

**d)** $8 - x \leq 3$ **e)** $12 - 3x > 6$ **f)** $-5x - 2 \geq 8$

**g)** $9 - 2x < 15$ **h)** $-4x + 1 \leq -11$ **i)** $2 - 5x \geq 12$

## E Both sides and fractions ●●●

**7.** Solve.

**a)** $6x - 5 \geq 2x + 7$ **b)** $8x + 3 < 5x - 6$ **c)** $5x + 4 > 2x + 10$

**d)** $3x + 11 \leq 7x + 3$ **e)** $\frac{x + 3}{2} < 6$ **f)** $\frac{2x - 3}{5} \geq 3$

**g)** $\frac{3x + 1}{2} > 5$ **h)** $2x - 9 > 5x + 6$ **i)** $\frac{4x - 2}{3} \leq 6$

## ★ Extension

**8.** Solve $-1 < 3x + 2 \leq 11$. Then list the whole numbers that satisfy it.

**9.** Movie tickets cost \$16 each and a large popcorn costs \$6. Hemi has \$70 and buys one popcorn. Write an inequality for the number of tickets $x$ he can buy, and solve it. What is the largest number of tickets?


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